

Static and Dynamic System

①

→ static $\Rightarrow$ the system depend only present time.

E.X 1 ① $y(t) = x(t+1) + x(t)$

③ $y(t) = x(2t)$

② $y(t) = e^{-(t+1)} \cdot x(t)$

④ a) $y(t-2) = x(t-2)$

b) $y(t+1) = x(t+1)$

Sol

① $y(t) = x(t+1) + x(t)$
 $\downarrow$ $\downarrow$
 Future Present

$\Rightarrow$ the system is Dynamic

② $y(t) = e^{-(t+1)} \cdot x(t)$
 coeff.

So the system depend on $x(t)$

the input signal [$x(t) \Rightarrow$ present]

the system is static.

③ $y(t) = x(2t)$

$t \Rightarrow 0 \quad y(0) = x(0)$

$t \Rightarrow 1 \Rightarrow y(1) = x(2)$

$t \Rightarrow 2 \quad y(2) = x(4)$

the system is Dynamic.

④ a) $y(t-2) = x(t-2)$

b) $y(t+1) = x(t+1)$

a) the system is static because $t \Rightarrow 0 \quad y(-2) = x(-2)$

$t \Rightarrow 1 \quad y(-1) = x(-1)$

the past value of output sys depend on past value of input sys

b) $y(t+1) = x(t+1)$ the system is static

$t \Rightarrow 0 \quad y(1) = x(1)$ the future output depend on future input for the same instant.

Ex 2 ① $y(t) = x(t)$

② $y(t) = \int_{-\infty}^t x(\tau) \cdot d\tau$

③ $y(t) = x(\sin(t))$

②

test the system static or dynamic?

Sol:

① $y(t) = x(-t)$

$t=0 \Rightarrow y(0) = x(0)$

$t=1 \Rightarrow y(1) = x(-1)$

$t=-1 \Rightarrow y(-1) = x(1)$

the system is Dynamic

② $y(t) = \int_{-\infty}^t x(\tau) \cdot d\tau$

to compute the Area we $t=0$ to know the value of $x(t)$ from $t=-\infty$ to $t=0$ that mean the system is Dynamic.

③ $y(t) = x(\sin(t))$

$t=0 \Rightarrow y(0) = x(\sin(0))$
 $= x(0)$

$t=\pi \Rightarrow y(\pi) = x(\sin(\pi))$
 $= x(0)$
 $y(3.14) = x(0)$

the system is Dynamic

H.W test the systems if static or Dynamic?

① $x(t+2) + 20 = y_1(t)$

② $x(t+3) \cdot \sin(2t) = y_2(t)$

③ $3x(2t) = y_3(t)$

Ex 3 Test the static and Dynamic for the system?

(3)

① $y(t) = x(t) + x(t^2)$

② $y(t) = \frac{d}{dt} x(t)$

Sol 8

① $y(t) = x(t) + x(t^2)$

$t=0 \Rightarrow y(0) = x(0) + x(0)$

$t=1 \Rightarrow y(1) = x(1) + x(1)$

$t=2 \Rightarrow y(2) = x(2) + x(4)$ the system is Dynamic

② $y(t) = \frac{d}{dt} x(t)$

to compute the change in $x(t)$
we need another point in future
or in Past and that make the system
is dynamic.

„Causal and non Causal system“

non Causal system $\rightarrow$ depend on Future time.

EX test if the system is causal or non causal?

① $y(t) = x(2t)$

③ $y(t) = \sin(t+1) x(t-1)$

② $y(t) = \begin{cases} x(5t) & t \leq 0 \\ x(t-3) & t \geq 0 \end{cases}$

④ $y(t) = x(e^t)$

Sol 8

① $y(t=0) = x(0)$

$y(t=1) = x(2)$

$y(t=1) = x(2)$

② for $t \leq 0$

$y(t) = x(5t) \quad t=-1 \quad y(-1) = x(-5)$

$y(-2) = x(-10)$

$y(-3) = x(-15)$

for $t \geq 0$

$y(t) = x(t-3)$

$t=0 \quad y(0) = x(-3) \quad ; \quad t=1 \quad y(1) = x(-2)$

$t=1 \quad y(1) = x(-2) \quad$ the system is Causal.

the system

is non-causal

③ $y(t) = \sin(t+1) \cdot x(t-1)$

$\sin(t+1)$ is the coeff so $y(t)$ depend on $x(t-1)$ is Past
the system is Causal system.

④ $y(t) = x(e^t)$ note: e^t is not coeff.

$t=0 \Rightarrow y(0) = x(e^0) \Rightarrow y(0) = x(1)$

the system is non causal.

Ex2 Test the following systems if they Causal or non Causal?

① $\int_{-\infty}^t x(\tau) \cdot d\tau = y(t)$

② $\int_{-\infty}^{t+1} x(\tau) \cdot d\tau = y(t)$

③ $\int_{-\infty}^t x(2\tau) \cdot d\tau = y(t)$

④ $\frac{d}{dt} x(t) = y(t)$

Sol In General in integration we need two Point to compute the result :

① $\int_{-\infty}^t x(\tau) \cdot d\tau \Rightarrow \begin{matrix} x(t) \\ x(-\infty) \end{matrix}$ the system is Causal

② $\int_{-\infty}^{t+1} x(\tau) \cdot d\tau \Rightarrow \begin{matrix} x(t+1) \\ x(-\infty) \end{matrix}$ the system is non-Causal.

③ $\int_{-\infty}^t x(2\tau) \cdot d\tau \Rightarrow \begin{matrix} x(2t) \\ x(-\infty) \end{matrix}$ the system is non causal

④ $\frac{d}{dt} x(t)$ also we need two Point and Present Point

↖
↗

Present Point + Past Point
the system is Causal

Present Point + Future Point
the system is non Causal

* the real physical system is Causal so the is Causal.

Q10 ① $y(t) = \int_{-\infty}^{3t} x(\tau/3) d\tau$

② $y(t) = \int_{-\infty}^t x(\tau-3) d(\tau-3)$

③ $y(t) = \int_{-\infty}^{3t} x(2\tau) d\tau$

Test if
the following systems
are causal or non causal?

• Time invariant and Time variant systems ..

the steps for solution

step ① compute $y(t-K)$ from $y(t)$ by change $t \rightarrow t-K$

step ② compute $y(t,K)$ from change $x(t) \rightarrow x(t-K)$ and compare

$y(t,K)$ and $y(t-K)$, if they equal the system is Time invariant
if not the system is Time variant.

E.x Test if the system is Time invariant or Time variant

① $y(t) = x(t) + 5$

② $y(t) = x(5t)$

Soln

① $y(t-K) = x(t-K) + 5$

$y(t,K) = x(t-K) + 5$

the system is Time invariant

② $y(t-K) = x(5(t-K))$

$= x(5t-5K)$

$y(t,K) = x(5t-K)$ the input to system is $x(t-K)$

only the scaling of $t \Rightarrow y(t,K) = x(5t-K)$

So $y(t-K) \neq y(t,K)$ the system is Time variant system

Note the system is Time Invariant system if

⑥

- ① no time scaling
- ② G_{eff} : should be constant.
- ③ Any added/subtracted term in the system relationship (except i/p and o/p) must be constant or zero.

E.x

- ① $y(t) = x(\cos(t))$
- ② $y(t) = x(t^2)$
- ③ $y(t) = \cos(t) \cdot x(t)$
- ④ $y(t) = e^{-t} \cdot x(t)$
- ⑤ $y(t) = 2t + x(t)$

sol all these systems are Time Variant by help the above notes.

H.w

- ① $y(t) = x(\log t)$
- ② $y(t) = t^2 \cdot x(t)$
- ③ $y(t) = 2\log(t) - x(t)$

E.x test the system if Time invariant or Time Variant?

① $y(t) = \int_{-\infty}^t x(\tau) \cdot d\tau$ $x(t) \rightarrow$ the system is Time Invariant system.

② $y(t) = \int_{-\infty}^t \cos(\tau) x(\tau) d\tau$ $\cos(t) \cdot x(t)$ the system is Time Variant system.

Notes any split system is Time Variant system because the coeff don't constant

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Ex 1:

$$y(t) = \begin{cases} x(t-1) & , t < 0 \\ x(t+1) & , t \geq 0 \end{cases} \quad \text{test if } y(t) \text{ Time Variant or Time invariant sys?}$$

Sol IN fact

$$y(t) = a(t)y(t-1) + b(t)y(t+1) \quad \text{where} \quad \begin{cases} a(t) = 1 & t < 0 \\ b(t) = 1 & t \geq 0 \end{cases}$$

So $a(t) \neq b(t)$ the coeff don't constant

that makes the system $y(t)$: Time Variant system.

Ex 2

$$y(t) = \begin{cases} x(t-2) & , t < -3 \\ x(t) & , -3 \leq t \leq 2 \\ x(t+2) & , t > 2 \end{cases} \quad \text{test if } y(t) \text{ Time Variant or Time invariant system?}$$

Sol IN fact

$$y(t) = a(t)x(t-2) + b(t)x(t) + c(t)x(t+2) \quad \text{where}$$

$$a(t) = 1 \quad t < -3$$

$$b(t) = 1 \quad -3 \leq t \leq 2$$

$$c(t) = 1 \quad t > 2$$

because of the coeff don't equal to constant that makes

$y(t)$ Time Variant system.

"Linear and non Linear Systems"

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Ex 1 : $y(t) = x(\sin t)$ test the Linearity of $y(t)$?

Sol: if input $= a_1 x_1(t) \Rightarrow y_1(t) = a_1 x_1(\sin t)$

" " $= a_2 x_2(t) \Rightarrow y_2(t) = a_2 x_2(\sin t)$

$$y_1(t) + y_2(t) = a_1 x_1(\sin t) + a_2 x_2(\sin t)$$

if input $= a_1 x_1(t) + a_2 x_2(t)$

$$\text{the output} = a_1 x_1(\sin t) + a_2 x_2(\sin t)$$

the system is Linear.

Ex 2 : $y(t) = x(t^2)$ Test the Linearity of $y(t)$?

Sol: if input $= a_1 x_1(t) \Rightarrow$ the output $= y_1(t) = a_1 x_1(t^2)$

" " $= a_2 x_2(t) \Rightarrow$ " " " $= y_2(t) = a_2 x_2(t^2)$

$$y_1(t) + y_2(t) = a_1 x_1(t^2) + a_2 x_2(t^2)$$

if the input $= a_1 x_1(t) + a_2 x_2(t)$ Then the output $= a_1 x_1(t^2) + a_2 x_2(t^2)$

So the system is Linear

note 1: From above we can conclude the time scaling don't effect on the Linearity of system.

H.W $y(t) = x(3t)$?

$y(t) = x(e^{3t})$?

Test the Linearity of the system?

Ex3: $y(t) = x^2(t)$ Test the Linearity of the system?

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Sol s=

if the input $\Rightarrow q_1 x_1(t) \Rightarrow$ the out put $y_1(t) = a_1 x_1^2(t)$

" " $\Rightarrow q_2 x_2(t) \Rightarrow$ " " $y_2(t) = a_2 x_2^2(t)$

$$y_1(t) + y_2(t) = a_1 x_1^2(t) + a_2 x_2^2(t)$$

if the input $\Rightarrow a_1 x_1(t) + a_2 x_2(t)$ the out put

$$= (a_1 x_1(t) + a_2 x_2(t))^2$$

So the system is non Linear.

note 28 from above, we can conclude the Power of system makes the system non-Linear.

H.w $y(t) = x(3t)^3$ Test the Linearity of $y(t)$?

Ex4 s ① $y(t) = \sin(t) \cdot x(t)$
② $y(t) = e^t \cdot x(t)$ test the Linearity?

Sol s ① If the input $= a_1 x_1(t)$ the out put $= y_1(t) = a_1 \sin(t) x_1(t)$

" " $= a_2 x_2(t)$ " " $= y_2(t) = a_2 \sin(t) x_2(t)$

$$y_1(t) + y_2(t) = a_1 \sin(t) x_1(t) + a_2 \sin(t) x_2(t)$$

If the input $= a_1 x_1(t) + a_2 x_2(t)$ then the out put

$$\text{equals to } \sin t [a_1 x_1(t) + a_2 x_2(t)]$$

$$= a_1 \sin(t) x_1(t) + a_2 \sin(t) x_2(t)$$

So the system is Linear

$$\textcircled{2} y(t) = e^5 X(t)$$

If the input signal = $a_1 X_1(t)$; the out put = $y_1(t) = e^5 \cdot a_1 X_1(t)$

" " " " = $a_2 X_2(t)$; " " " " = $y_2(t) = e^5 \cdot a_2 X_2(t)$

$$y_1(t) + y_2(t) = e^5 a_1 X_1(t) + e^5 a_2 X_2(t)$$

$$= e^5 (a_1 X_1(t) + a_2 X_2(t))$$

If the input signal = $a_1 X_1(t) + a_2 X_2(t)$; the the out put

$$\text{equal to } e^5 [a_1 X_1(t) + a_2 X_2(t)]$$

So the system is Linear.

note 3: from above we can conclude, the constant coefficient independent coefficient don't effect on the Linearity of systems.

H.W test the Linearity of

$$\textcircled{1} y(t) = \log(t) X(t) + 3t^2 X(3t)$$

Ex 5 test the Linearity for $\textcircled{1} y(t) = 4t + X(t)$

$$\textcircled{2} y(t) = 5 + X(t)$$

Sol 8

If the input = $a_1 X_1(t)$; the out put = $y_1(t) = 4t + a_1 X_1(t)$

" " " " = $a_2 X_2(t)$; " " " " = $y_2(t) = 4t + a_2 X_2(t)$

$$y_1(t) + y_2(t) = 4t + a_1 X_1(t) + 4t + a_2 X_2(t)$$

$$= 8t + a_1 X_1(t) + a_2 X_2(t)$$

If the input = $a_1 X_1(t) + a_2 X_2(t)$; then the out put

$$\text{equal to } 4t + a_1 X_1(t) + a_2 X_2(t)$$

So the system is Non Linear.

(2) If the input signal = $a_1 x_1(t)$; the output = $y_1(t) = 5 + a_1 x_1(t)$
 $\quad \quad \quad \quad \quad \quad = a_2 x_2(t)$; $\therefore y_2(t) = 5 + a_2 x_2(t)$

$$y_1(t) + y_2(t) = 5 + a_1 x_1(t) + 5 + a_2 x_2(t)$$

$$= 10 + a_1 x_1(t) + a_2 x_2(t)$$

If the input signal $= a_1 x_1(t) + a_2 x_2(t)$; the output equal to

$$5 + a_1 x_1(t) + a_2 x_2(t)$$

So the system is non Linear.

note, 4s from above, we can conclude, the dc shifting on system or dependant Adding [term constant t] make the system non Linear.

H.W $y(t) = 3\log(2t) + \cos(2t) \times (5t)$ Test the Linearity?

E.X6 8- $y(t) = x(t+1) + x(t-2)$ Test the Linearity of system?

Sol 8 If the Input = $a_1 x_1(t)$ the output = $y_1(t) = a_1 x_1(t+5) + a_1 x_1(t-2)$
 $\therefore \quad \quad \quad = a_2 x_2(t) \quad \therefore \quad \quad = y_2(t) = a_2 x_2(t+5) + a_2 x_2(t-2)$

$$y_1(t) + y_2(t) = a_1 x_1(t+5) + a_2 x_2(t+5) + a_1 x_1(t-2) + a_2 x_2(t-2)$$

If the input $= a_1 x_1(t) + a_2 x_2(t)$; then the output

$$a_1 x_1(t+5) + a_1 x_1(t-2) + a_2 x_2(t+5) + a_2 x_2(t-2)$$

So the system is Linear.

note 5: From above we can conclude the time shift try don't effect on Linearity of the system.

H.W Test the Linearity of the system

$$y(t) = \sin 2t \cdot x(t-5) + e^{2t} x(t^2)$$

E.X 7 :- ① $y(t) = \int_{-\infty}^t x(\tau) \cdot d\tau$ ② $y(t) = \frac{d}{dt} x(t)$

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Test the systems Linearity 2.

Soln

① input = $a_1 x_1(t)$ $\Rightarrow$ output = $y_1(t) = a_1 \int_{-\infty}^t x_1(\tau) d\tau$
 $\therefore = a_2 x_2(t) \Rightarrow \quad \quad \quad = y_2(t) = a_2 \int_{-\infty}^t x_2(\tau) \cdot d\tau$

$$y_1(t) + y_2(t) = a_1 \int_{-\infty}^t x_1(\tau) d\tau + a_2 \int_{-\infty}^t x_2(\tau) \cdot d\tau$$

$$= \int_{-\infty}^t [a_1 x_1(\tau) + a_2 x_2(\tau)] d\tau$$

input = $a_1 x_1(t) + a_2 x_2(t)$; output = $\int_{-\infty}^t [a_1 x_1(\tau) + a_2 x_2(\tau)] d\tau$

the system is Linear.

② $y(t) = \frac{d}{dt} x(t)$?

input = $a_1 x_1(t)$; output = $\frac{d}{dt} a_1 x_1(t) = y_1(t)$

$\therefore = a_2 x_2(t) ; \quad \quad \quad = \frac{d}{dt} a_2 x_2(t) = y_2(t)$

$$y_1(t) + y_2(t) = \frac{d}{dt} a_1 x_1(t) + \frac{d}{dt} a_2 x_2(t)$$

$$= \frac{d}{dt} [a_1 x_1(t) + a_2 x_2(t)]$$

input = $a_1 x_1(t) + a_2 x_2(t)$; the output = $\frac{d}{dt} [a_1 x_1(t) + a_2 x_2(t)]$

So the system is Linear.

note 68 From above, Both integration and differential operator don't effect on Linearity of the system.

H.W $y(t) = 5 \int_{-\infty}^t x(\tau) d\tau + \frac{d}{dt} x(t^3)$ test the Linearity ?

note 780 All these operations are non Linear — ① Even and odd operation; ② Real and imaginary operation; ③ Trigonometric and inverse trigonometric operation; ④ Logarithmic and Exponential operation; ⑤ Power and roots operations; ⑥ modulus; ⑦ sgn, Sampling; sinc, and other —

note 880 Fast testing for Linearity of system when input signal = Zero the output signal equal to Zero For Linear system.

Ex 880 test the Linearity For ① $y(t) = 2x(t)$
② $y(t) = 2 + x(t)$

Sol ①
For $x(t) = 0 \Rightarrow y(t) = 2 \cdot 0 \Rightarrow \text{out put} = \text{Zero}$ the sys is Linear.

②
for $x(t) = 0 \Rightarrow y(t) = 2 + 0 \Rightarrow \text{out put} = 2$ the sys is non Linear.

Ex 900 test the Linearity for $y(t) = \begin{cases} 2x(t+\pi) & t < 0 \\ x(t^2) & t \geq 0 \end{cases}$

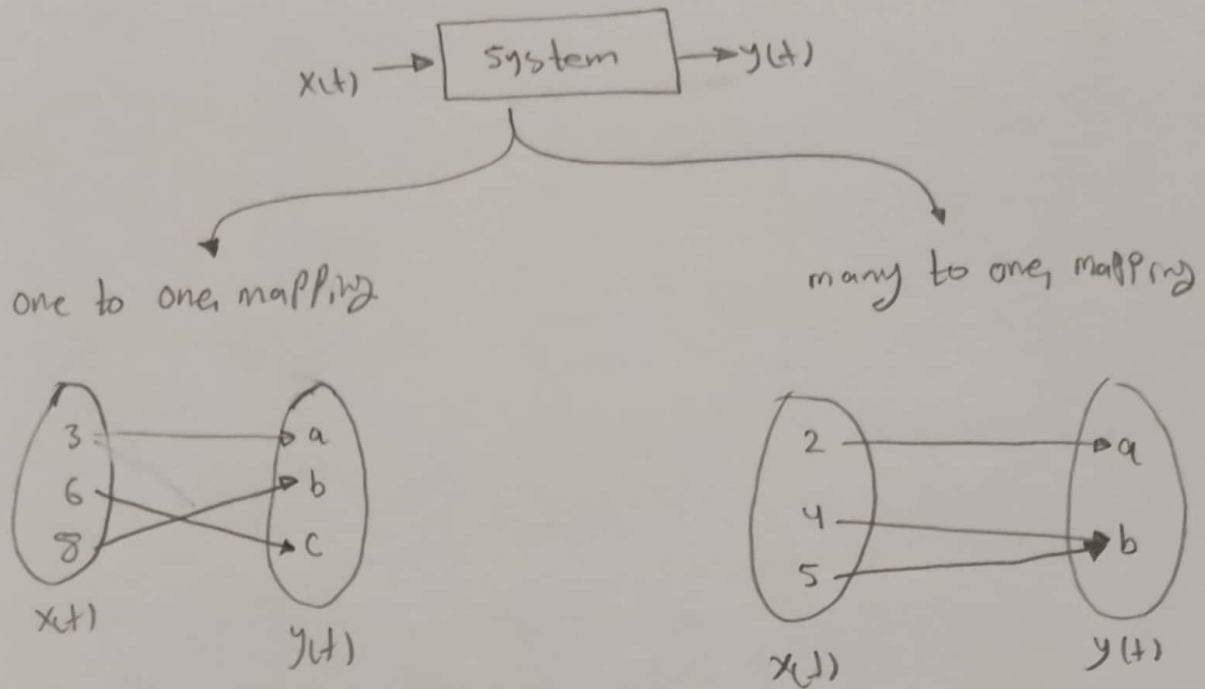
Sol $y(t) = a(t) \cdot 2x(t+\pi) + b(t) \cdot x(t^2)$ when $\begin{cases} a(t) = 1 & t < 0 \\ b(t) = 1 & t \geq 0 \end{cases}$

and $a(t)$ and $b(t)$ don't effect on Linearity of system

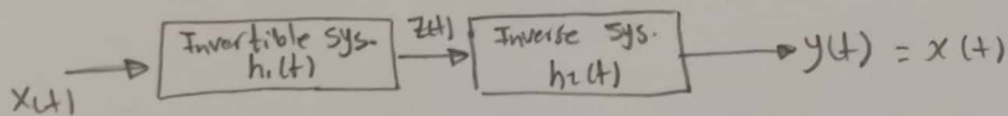
$2x(t+\pi) \Rightarrow \text{Linear}$ $x(t^2) : \text{Linear}$

so $y(t) = \text{Linear system}$.

note 980 split system maybe Linear but Not Time invariant system.



→ for an invertible system, there should be one to one, between I/P and O/P at each and every instant of time.



$$x(t) \rightarrow [h_1(t) * h_2(t)] \rightarrow y(t) = x(t)$$

$$h_{eq} = h_1(t) * h_2(t)$$

L.T ↓

$$H_{eq}(s) = H_1(s) \cdot H_2(s) \quad \text{where } H_2(s) = \frac{1}{H_1(s)}$$

$$= H_1(s) \cdot \frac{1}{H_1(s)} \Rightarrow H_{eq}(s) = 1$$

→ another description for invertible system by using

$$S\text{-Plane} \Rightarrow H_{eq}(s) = 1.$$

Ex Test if the system is invertible, or Non-invertible?

① $y(t) = x^2(t)$

② $y(t) = x(t) + 2$

Sol ① $y(t) = x^2(t)$

$x(t)$	$y(t)$
1	1
-1	1
2	4
-2	4

many to one mapping
the system is
Non-invertible

② $y(t) = x(t) + 2$

$x(t)$	$y(t)$
0	2
1	3
-1	1
2	4
-2	0

one to one mapping
the system is invertible

Ex2 Test if the system is invertible, or non invertible?

① $y(t) = |x(t)|$

② $y(t) = \sin(t) x(t)$

$x(t)$	$y(t)$
1	1
-1	1
2	2
-2	2

many to one mapping
the system is noninvertible

$x(t)$	$y(t) = \sin(t) x(t)$
0	$y(t) = \sin(t) \cdot 0 = 0$
1	$y(t) = \sin(t)$
2	$y(t) = 2 \sin(t)$
$\delta(t)$	$y(t) = \sin(t) \cdot \delta(t - 0)$ $= \sin(0) \cdot \delta(t)$ $= 0$
$2\delta(t)$	$y(t) = \sin(t) \cdot 2\delta(t)$ $= 2 \sin(0)$ $= 0$

many to one mapping
the sys is non-invertible.

③ $y(t) = x(2t)$

$x(t)$	$y(t)$
$u(t)$	$u(2t)$
$-u(t)$	$-u(2t)$

one to one mapping
the sys is invertible.

H.w test the invertibility?

① $y(t) = x(t^2)$

② $y(t) = \frac{d}{dt} x(t)$

"Stable and unstable systems"

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Boundary i/p Boundary o/p Criteria = for a stable system o/p should be bounded for bounded i/p at each and every instant of time.

→ Bounded signal : dc, $\sin(t)$, $\cos(t)$, $u(t)$
 $-1 \text{ to } 1$ $-1 \text{ to } 1$ $0 \text{ or } 1$

Ex 1: ① $y(t) = t \cdot x(t)$

② $y(t) = x(t) + 2$

Test the stability of the system?

Sol

① $x(t) = u(t)$ $y(t) = t \cdot u(t)$
 $= r(t)$

the system is unstable.

② $x(t) = dc = 4$ $y(t) = 4 + 2$

$= 6 \text{ dc}$ the system is stable.

Ex 2: ① $y(t) = \sin[x(t)]$ ② $y(t) = \int_{-\infty}^t x(\tau) \cdot d\tau$

Test the stability of the system?

Sol

① $y(t) = \sin[x(t)]$

if $x(t) = u(t)$

$y(t) = \sin(0 \text{ or } 1)$

and $y(t)$: Bounded

for $x(t) = \cos(t) \Rightarrow -1 \text{ or } 1$

$y(t) = \sin[+1 \text{ or } -1]$

$y(t)$: Bounded

the system is stable.

② $y(t) = \int_{-\infty}^t x(\tau) \cdot d\tau$

for $x(t) = \cos t$

$y(t) = \int_{-\infty}^t \cos \tau \cdot d\tau = [\sin \tau]_{-\infty}^t$

$= \sin(t) - \sin(-\infty)$

$= \sin(t) + \sin(\infty) \Rightarrow y(t) : \text{finite}$

= stable system

But for $x(t) = u(t)$ t

$y(t) = \int_{-\infty}^t x(\tau) \cdot d\tau = \int_{-\infty}^t u(\tau) \cdot d\tau$

$= r(t)$ the system is unstable.